

2.2 Job Listing Data

The second Revelio Labs dataset contains almost the entire universe of online job postings in the UK over the same period, 63,497,648 in total. The data are collected from vacancies from major job boards, company career pages, and recruitment platforms. Each posting includes the job title, employer, posting and removal dates, salary information, and full job ads. This job listing data captures labor demand flows and allows us to observe how hiring intentions and job characteristics evolve. We use this dataset for two purposes: to examine how hiring intentions evolve across exposed and less-exposed firms and across exposed and less-exposed occupations.

To construct our firm-level hiring intentions data, we use the same sample of firms with at least 100 employees in October 2022 used in our total employment analysis, but focus on flows (job listings) instead of employment stocks. We create a monthly panel at the firm level, tracking job posting activity from January 2021 through June 2025. Given that approximately 50% of firms do not post job listings in any given month, we construct a binary outcome variable indicating whether a firm posted any job listings in a given month, rather than using posting counts. This measure captures the extensive margin of hiring activity. We again match each firm to its pre-treatment LLM exposure score calculated from the October 2022 workforce composition, as described above. We also break postings down by job roles (50 in total) to study which types of roles exposed firms hire more or less of.

This firm-level posting dataset allows us to test whether firms with higher LLM exposure adjust their hiring intensity differently than less-exposed firms following ChatGPT’s introduction, providing insights into how anticipated productivity gains or workforce restructuring needs translate into recruitment behavior. In addition, by examining the occupational composition of job postings within firms, we can identify whether highly exposed firms disproportionately reduce hiring for specific types of roles (for example, those with high LLM exposure scores) relative to less exposed firms, allowing us to test for within-firm occupational reallocation in response to LLM adoption.

To construct our occupation-level hiring intentions data, we first match each vacancy to O*NET SOC occupation codes using a job-role-SOC crosswalk provided by Revelio Labs. This crosswalk maps each unique role (1,500 in total) to 384 broader categories that directly correspond to the occupations listed in Eloundou et al.

Table 2: Summary Statistics Occupation-Level Job Postings

	Low exposure		High exposure	
	Before	After	Before	After
Job postings	4,056	3,358	3,343	2,745
Salary	£44,828.34	£49,359.93	£46,177.33	£50,262.62
Salary (SD)	£13,940.86	£15,238.06	£13,469.71	£14,136.39
LLM exposure score	0.045	0.045	0.263	0.263

Notes: This table reports occupation-level summary statistics on job postings by exposure to large language models (LLMs). Occupations are classified into high- and low-exposure groups based on whether their exposure score lies above or below the median score. *Job postings* gives the mean number of vacancies posted. *Salary* reports the mean advertised annual compensation, while *Salary (SD)* gives the corresponding standard deviation across postings. *LLM exposure score* is the average predicted LLM-exposure of tasks in the occupation, based on Eloundou et al. (2024). *Before* refers to the pre-treatment period (prior to November 2022), while *After* refers to the post-treatment period (November 2022 onward).

(2024). We then aggregate vacancies to the occupation-month level. Our monthly occupation-level variables capture multiple dimensions of labor demand. Posting volume is measured as the total new vacancies posted, which we log-transform. Posted salary represents the mean advertised salary for positions (level or log).⁷ We additionally calculate salary dispersion as the standard deviation of posted salaries within occupation-month cells to capture within-occupation inequality effects.

Table 2 presents summary statistics for occupation-level job postings. We separate the data by high vs. low exposure occupations, and show the data for the periods before and after the release of ChatGPT in November 2022. Job posting volumes declined across both exposure groups following the introduction of ChatGPT, with high-exposure occupations experiencing a sharper decline of 17.9 percent (from 3,343 to 2,745 monthly postings) compared to 17.2 percent for low-exposure occupations (from 4,056 to 3,358). Second, average posted salaries increased for both groups, rising from £44,828 to £49,360 (10.1 percent) in low-exposure occupations and from £46,177 to £50,263 (8.8 percent) in high-exposure occupations, with salary dispersion increasing similarly across both groups. Finally, high-exposure occupations have LLM exposure scores approximately 6 times higher than low-exposure occupations.

⁷When salary is not explicitly listed, it is inferred from a predictive model based on job characteristics, company, and location, trained on visa applications, self-reported data, and job postings, with salaries inferred for companies lacking direct data.

3 Methodology

We use a difference-in-differences design with continuous treatment intensity to estimate the effect of LLM exposure on UK labor market outcomes. Our identification strategy exploits variation in occupational and firm-level exposure to LLM capabilities, as measured by Eloundou et al. (2024), combined with the timing of ChatGPT’s public release in November 2022. We conduct three types of analyses: firm-level employment stocks using LinkedIn profile data, firm-level hiring decisions based on job posting data, and occupation-level hiring patterns using job posting data.

For our firm-level employment stock analysis, we estimate the following specification:

$$Y_{it} = \sum_{\tau=-15}^{30} \beta_{\tau} \cdot \text{Exposure}_i \times \mathbb{1}_{[t=\tau]} + \gamma_i + \delta_{jt} + \epsilon_{it} \quad (3)$$

where Y_{it} represents outcomes for firm i in month t , including log total employment, log employment by seniority level (junior vs. senior), the fraction of low-seniority positions, and average compensation. Exposure_i is the firm’s pre-treatment LLM exposure score, calculated as the employment-weighted average of occupational exposure scores using the workforce composition on 30 October. We standardize the exposure coefficients across all analyses to make the coefficients more easily interpretable. The term $\mathbb{1}_{[t=\tau]}$ denotes indicators for event time τ relative to November 2022, with October 2022 ($\tau = -1$) as the reference period. We include firm fixed effects γ_i to control for time-invariant firm characteristics, and industry-by-period fixed effects δ_{jt} to account for sector-specific shocks that might correlate with technology adoption patterns. Standard errors are clustered at the firm level.

This event study design allows us to test for parallel pre-treatment trends, as well as to examine the evolution of treatment effects over time, showing when and how labor market impacts start to emerge, and whether they persist, intensify, or attenuate over time. We use 15 months of pre-treatment data and 30 months of post-treatment data. To summarize the overall treatment effect, we also report average post-treatment effects from a static specification:

$$Y_{it} = \beta \cdot \text{Exposure}_i \times \text{Post}_t + \gamma_i + \delta_{jt} + \epsilon_{it} \quad (4)$$

where Post_t is an indicator for periods after November 2022.

To examine how firms adjust their hiring strategies in response to LLM exposure, we combine our firm-level exposure measures with job posting data. To do so, we estimate Equation (3) with an indicator for whether firm i posted any job listings in month t as the outcome variable. This binary outcome captures the extensive margin of total hiring activity. We then re-estimate Equation (4) separately for each of 50 job categories.

For our occupation-level analysis of job posting dynamics, we estimate an analogous event study specification:

$$Y_{ot} = \sum_{\tau=-15}^{30} \beta_{\tau} \cdot \text{Exposure}_o \times \mathbb{1}_{[t=\tau]} + \gamma_o + \delta_t + \epsilon_{ot} \quad (5)$$

where Y_{ot} denotes outcomes for occupation o in month t , including log job posting volumes, posted salaries, and salary dispersion within occupations. Exposure_o is the occupation’s LLM exposure score from Eloundou et al. (2024). We include occupation fixed effects γ_o and year-by-month fixed effects δ_t with standard errors clustered at the occupation level. We also estimate an analogous static specification of Equation (4) at the occupation level.

Our identification strategy relies on the parallel trends assumption: absent ChatGPT’s introduction, labor market outcomes would have evolved similarly across firms and occupations with different LLM exposure levels, conditional on our fixed effects structure. Several features of our design help with the interpretation of our estimates as causal. First, our exposure measures are predetermined using data from before November 2022, eliminating concerns about reverse causality where employment changes might influence measured exposure. Second, the sudden and largely unanticipated public release of ChatGPT provides a clear temporal shock to LLM availability. Third, our event study analyses test and confirm parallel pre-treatment trends across analyses.

Furthermore, we perform a large set of robustness checks to test the validity of our findings across different specifications and modeling choices. We re-estimate our analyses using different fixed effects. We examine the validity of our finding of decreased junior hiring through a highly restrictive within-firm triple-difference model using a firm-by-month-by-seniority panel that includes firm-by-period fixed effects absorbing all time-varying firm-level shocks. We test the sensitivity of our results to